🛠️

26-SUPPORT-RUNBOOK-LIBRARY.md — One page per error code, written before the ticket arrives

> 🛠️ Every error code in `11-ERROR-TAXONOMY.md` must have a runbook before the code is allowed to exist. A code without a runbook is a support ticket you have already decided to handle badly at 2 a.m.

1. The rule

CI fails if any code registered in `errors.toml` lacks `docs/runbooks/<code>.md`. This is a hard gate, not a documentation aspiration. The runbook is written by the engineer who introduced the code, at the moment they introduce it, while they still remember what causes it.

2. Runbook template

```
# AURA-RAW-2008 — RAW decode produced implausible dimensions

**Severity:** ItemFailed  **Recovery:** Quarantine  **Introduced:** v0.4.0

## What the user sees
"3 photos could not be read and were set aside. Everything else finished normally."

```

```
## What actually happened
LibRaw returned a frame whose dimensions disagree with the EXIF header,
usually a truncated file or an unsupported compressed variant.

## First three questions to ask
1. Which camera model? (most 2008 cases are one specific body)
2. Was the card ejected during copy, or copied over a network?
3. Does the file open in the camera manufacturer's own software?

## Diagnosis
- Ask for the run bundle (Help › Export diagnostics). Open `errors.json`.
- Check `quarantine/` for the affected files' metadata sidecars.
- Reproduce: `cargo run -p aura-cli -- decode-probe <file>`

## Resolution
| Cause | Fix | Permanent? |
|---|---|---|
| Truncated file | Re-copy from card | User-side |
| Unsupported variant | Add camera to compatibility list; ship LibRaw bump | Needs release |
| Genuinely corrupt | Nothing recoverable; confirm quarantine behaved | N/A |

## Escalate when
More than 2 % of a single import hits this code, or the camera is
already on the supported list.

## Prevention
```

```
FMEA row F-11. Guard: dimension plausibility check at decode
boundary.
```

3. Coverage by domain

Domain	Runbooks needed	Most common real-world ticket
IO 1000–1999	~18	External drive disconnected mid-import; long Windows paths; network volumes
RAW 2000–2999	~14	A camera body released after your last LibRaw bump
DB 3000–3999	~9	Catalog on Dropbox or OneDrive — the single most destructive user habit
GPU 4000–4999	~12	Driver timeout under load; laptop switching to integrated graphics on battery
ML 5000–5999	~11	Model pack download incomplete; EP unavailable after a driver update
CLOUD 6000–6999	~8	Spend cap reached; captive-portal hotel wifi returning HTML
JOB 7000–7999	~10	Machine slept mid-run; resumed run appears stuck
EXPORT 8000–8999	~9	Disk full at 3,800 of 4,000; destination folder permissions
SEC 9000–9999	~6	Consent not granted, user does not understand why cloud features are greyed out

4. The five tickets you will actually get most

Write these as full customer-facing help articles, not just internal runbooks:

1. **"It's slower than you promised."** Almost always thermal throttling on a laptop, an external drive over USB 2, or an integrated GPU. Ship a one-click Diagnose Performance report that names the bottleneck instead of arguing.
2. **"It picked the wrong photos."** Not a bug — a calibration mismatch. Route to style learning (`28-STYLE-LEARNING-SPEC.md`) and to the confidence threshold slider. Capture the disagreement as training signal if consent allows.

3. **"My catalog is corrupted."** Cloud-synced catalog folder, 90 % of the time. Detect it at project creation and refuse politely with an explanation, rather than fixing it after the fact.
4. **"The colours look different from Lightroom."** True, and expected — different rendering. Needs an honest comparison document, not a defensive one.
5. **"It crashed and I lost my work."** Verify resume actually worked; if it did not, this is a `JOB` domain bug and a P1, because R2 says every run is resumable.

5. Support tooling that pays for itself

- **Help › Export diagnostics** produces the run bundle from `19-OBSERVABILITY-AND-FORENSICS.md`, already redacted, as a single zip the user can attach. One click, no instructions needed.
- **A code lookup page**: `aura.app/e/AURA-RAW-2008` renders the user-facing half of the runbook. Put the URL in the error dialog itself.
- **Self-diagnose panel**: GPU and EP detected, disk speed, free space, thermal state, catalog location warning, model pack integrity. Most tickets resolve here without a human.
- **Severity SLAs**: Fatal 4 hours, RunBlocking 1 business day, ItemFailed 3 days, Degraded/Warning next release.

6. Acceptance criteria

- [ ] Every code in `errors.toml` has a runbook file; CI enforces it.
- [ ] Every runbook has all eight template sections filled with real content.
- [ ] The five common tickets exist as public help articles.
- [ ] Every error dialog in the UI links to its code lookup URL.
- [ ] Export diagnostics is reachable in one click and passes the redaction test.
- [ ] Ten runbooks have been walked by someone who did not write them, and their confusion recorded and fixed.

7. Claude Code prompt

> Act as `DOC` (Documentation Lead) with `TLC` reviewing. Read `errors.toml` and generate `docs/runbooks/<code>.md` for every registered code using the section 2 template. Do not invent causes — for each code, trace the actual call sites that emit it and describe the real conditions, citing file:line.

> Then add the CI gate that fails when a code has no runbook, and prove it by registering a dummy code without a runbook, showing the failure, and removing it. Finally, wire every error dialog to the code lookup URL, and list any code whose user-facing message does not state a next action.